Chapter # 08 Magnetism

Briefly describe magnetic shielding.

1) Magnetic Shielding

Magnetic shielding is the method of **reducing or blocking magnetic field effects** using materials like soft iron or mu-metal.

👉 It works by providing an **easy path for magnetic field lines**, so they don't enter the protected region.

Example: protecting sensitive electronic devices.

What is induced magnetism? Briefly describe the induced magnetism methods like stroking, hammering, and heating.

2) Induced Magnetism

Induced magnetism is when a material becomes a magnet **due to the influence of an external magnetic field**.

Methods:

- **Stroking:** Rubbing with a magnet aligns domains → becomes magnet
- **Hammering:** Aligns domains in Earth's magnetic field
- **Heating:** High temperature destroys magnetism (randomizes domains)

What is a solenoid? How solenoid can function as a magnet? Write equation for the magnetic field of solenoid.

3) Solenoid

A solenoid is a **long coil of wire**.

It acts as a magnet because:

- Current produces magnetic field
- Fields combine → behave like a bar magnet (with N & S poles)

Magnetic Field:

$$B = \mu_0 n I$$

Where μ_0 is the permeability of free space, n is the number of turns per unit length, and I is the current.

Does magnetic field lines always emerge from north pole? Can they move from south pole to the north pole? Justify.

4) Magnetic Field Lines Direction

- Outside magnet: **North → South**
- Inside magnet: **South → North**

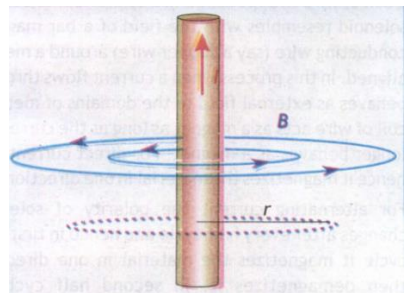
👉 So, field lines form **closed loops**, not one-way.

What happens if a current passed from a conductor? Write the equation of magnetic field in this case. Draw a diagram to support your answer.

When current passes through a straight conductor, a **circular magnetic field** is created around it. The direction is determined by the **Right-Hand Grip Rule**.

- **Equation:** The magnetic field B at a distance r from a long straight wire is:

$$B = \frac{\mu_0 I}{2\pi r}$$



Magnetic field around a wire

Differentiate between a permanent magnet and an electromagnet.

Permanent Magnet	Electromagnet
Always magnetic	Magnetic only with current
Fixed strength	Adjustable strength
No electricity needed	Requires electricity
Example: bar magnet	Example: solenoid

List some uses and applications of permanent magnet and electromagnet.

Permanent Magnet:

- **Compass**
- **Particle accelerators**
- **Transformers**
- **Speakers**

Electromagnet:

- **Electric Fans**
- **Electric Motors**
- **Doorbells & Door Locks**
- **MRI Machines**

Differentiate between diamagnetic , paramagnetic, and ferromagnetic materials with examples.

- **Ferromagnetic:** Strongly attracted to magnets; can become permanent magnets (e.g., Iron, Nickel, Cobalt).
- **Paramagnetic:** Weakly attracted to magnets; do not retain magnetism when the field is removed (e.g., Aluminum, Oxygen, Platinum).
- **Diamagnetic:** Weakly *repelled* by both poles of a magnet (e.g., Copper, Gold, Water, Bismuth).

Differentiate between magnetic and non-magnetic materials with examples.

- **Magnetic Materials:** Materials that can be magnetized or are attracted to magnets (e.g., Steel).
- **Non-Magnetic Materials:** Materials that show no visible reaction to magnetic fields (e.g., Wood, Plastic, Glass).

How does the earth create its own magnetic field?

How it's created: Earth's field is generated by the **Dynamo Effect**. This involves the rotation of the Earth combined with the convection of molten iron and nickel in the outer core. This moving conducting fluid creates massive electric currents, which in turn generate the magnetic field.

Explain the geographical and magnetic poles of the Earth?

Geographical Poles	Magnetic Poles
Based on Earth's rotation	Based on magnetic field
Fixed locations	Slightly shifting
True North/South	Magnetic North/South

What is meant by bio magnetism? Write down its applications.

Biomagnetism is the **magnetic field produced by living organisms**.

Applications:

- Brain studies (MRI, MEG)
- Heart activity measurement
- Medical diagnostics

Can two magnetic field lines intersect each other? Justify your answer.
A freely suspended magnet always points along north-south direction. Why?
What is the neutral zone or field free region of the magnetic field?
Is there any material which does not have any magnetic behavior? Justify your answer.

No, magnetic field lines **cannot intersect**.
👉 If they intersected, there would be **two directions of magnetic field at one point**, which is impossible because the magnetic field at a point has **only one definite direction**.

Because the **Earth behaves like a giant magnet**.
👉 Its magnetic field exerts a torque on the suspended magnet, aligning it along the **north-south direction**.

The neutral zone is a region where the **net magnetic field is zero**.
👉 It occurs when two magnetic fields (e.g., Earth's field and a magnet's field) **cancel each other**.

No material is completely non-magnetic.
👉 All materials show some magnetic effect:

- Diamagnetic (very weak repulsion)
- Paramagnetic (weak attraction)

However, some materials (like wood, plastic) appear **non-magnetic** because their effect is **extremely weak**.

What is the geomagnetic reversal phenomenon? Explain.
Why the Earth spins about its geographical axis instead of its magnetic axis? Explain.
Why the Earth's geographical and magnetic axis are not coincident? Explain.

Geomagnetic reversal is the phenomenon in which the **Earth's magnetic poles switch places** (North becomes South and vice versa).

👉 It happens due to changes in the **motion of molten iron in Earth's core**, which generates the magnetic field.
👉 This process occurs over **thousands of years** and has happened many times in Earth's history.

Earth rotates about its **geographical axis** because:

- It is determined by the **distribution of mass and rotation (angular momentum)** of the Earth.
- The **magnetic axis is not physical**; it is only related to the magnetic field.

👉 Therefore, rotation depends on **mass and motion**, not on magnetism.

The two axes are not the same because:

- The **geographical axis** is fixed by Earth's rotation.
- The **magnetic axis** is produced by **irregular motion of molten iron** inside Earth.

👉 Due to this, the magnetic axis is **tilted (~11°)** from the geographical axis and keeps changing slightly with time.

A student takes a long straight copper wire (with insulation on it) from his physics lab, he wraps the wire on an iron rod of 50 cm, to make it solenoid of 15 turns. He connects the ends of copper wire with the battery, which provides 1.2 A current. Find the magnetic field he produces in solenoid by doing so.

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Length 'L' = 50 cm = 0.5 m

Number of turns 'N' = 50

Current 'I' = 1.2 A

Permeability 'μ_o' = $4\pi \times 10^{-7}$ N/A²

REQUIRED

Magnetic Field Strength 'B' = ?

SOLUTION

To find magnetic Field Strength we have to find number of turns per unit length first by using equation 8.2

$$n = \frac{N}{L} = \frac{50}{0.5} = 100\,m^{-1}$$

Now using equation 8.1 $\quad B = \mu_o n I$

Putting values: $B = (4\pi \times 10^{-7} N A^{-2})(100\,m^{-1})(1.2A)$

Therefore, $\boxed{B = 1.5 \times 10^{-4}\ T}$ — Answer

Find the magnetic field due to a wire at 10 cm, if 1.3 A current is passing through the wire.

We use the magnetic field formula for a long straight current-carrying wire:

$$B = \frac{\mu_0 I}{2\pi r}$$

Given:

- $I = 1.3\,A$
- $r = 10\,cm = 0.1\,m$
- $\mu_0 = 4\pi \times 10^{-7}\,T\,m/A$

Solution:

$$B = \frac{(4\pi \times 10^{-7})(1.3)}{2\pi(0.1)}$$

Cancel π:

$$B = \frac{4 \times 10^{-7} \times 1.3}{2 \times 0.1}$$

$$B = \frac{5.2 \times 10^{-7}}{0.2}$$

$$B = 2.6 \times 10^{-6}\, T$$

Chapter # 9 Nature of Science

1) Explai briefly the following fields of Physics: Astrophysics, cosmology, optics, and astronomy.

- Astrophysics
  Study of physical properties of stars, planets, galaxies (e.g., temperature, motion, composition).
- Cosmology
  Deals with the origin, structure, and fate of the universe (e.g., Big Bang theory).
- Optics
  Study of light, reflection, refraction, lenses, and optical instruments.
- Astronomy
  Observation and study of stars, planets, moons, and galaxies.

2) Differentiate between classical, quantum and relativistic mechanics.

Classical Mechanics	Quantum Mechanics	Relativistic Mechanics
Deals with macroscopic objects	Deals with atomic/subatomic particles	Deals with very high speeds (near speed of light)
Based on Newton's laws	Based on probability & wave nature	Based on relativity (Einstein)
Exact results	Probabilistic results	Space & time are relative
Example: motion of car	Example: electron in atom	Example: fast-moving particles

3) Differentiate between hypothesis, theory, and law with examples from physics.

Hypothesis	Theory	Law
Tentative idea or assumption	Well-tested explanation	Universal rule
Needs testing	Supported by experiments	Always valid under given conditions
Example: “Light behaves as wave”	Atomic theory	Newton’s laws of motion

4) Differentiate between science, technology, and engineering.

Science	Technology	Engineering
Study of natural laws	Application of science	Designing & building systems
Focus on knowledge	Focus on tools/machines	Focus on problem-solving
Example: laws of motion	Mobile phones	Bridge construction